ICS445 Project Phase 1

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Introduction

This report documents the design, implementation, and configuration of a multi-site enterprise network infrastructure connecting two geographically distributed offices through secure WAN connectivity. The network architecture encompasses Office A and Office B, each equipped with redundant distribution layer switches (DSW-A1/A2 and DSW-B1/B2) and access layer switches (ASW-A1/A2/A3 and ASW-B1/B2/B3) to ensure high availability and scalability. Both sites feature wireless connectivity through lightweight access points (LWAP1 and LWAP2) managed by wireless LAN controllers (WLC), providing seamless mobility for end users. The core infrastructure is interconnected through core switches (CSW1 and CSW2) at the distribution layer, with dual WAN routers (R1) establishing redundant paths to the Internet for business continuity. Office B additionally hosts a server (SRV1) to support centralized services and applications. This hierarchical network design follows industry best practices, incorporating redundancy at multiple layers to eliminate single points of failure while maintaining optimal performance and security across the enterprise.

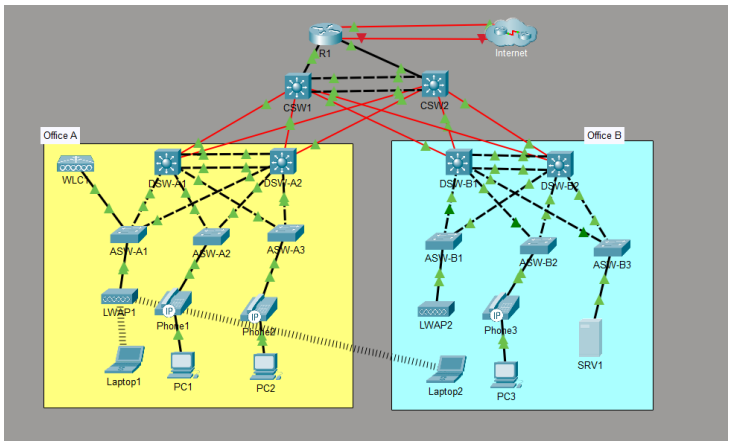

Figure 1: Whole Network

1 Part 1 - Initial Setup

This part covers essential security configurations for network devices, including setting unique hostnames, implementing encrypted enable secret and user account passwords (using type 9 or type 5 hashing), and securing console access with local authentication, a 30-minute inactivity timeout, and synchronous logging. These measures establish foundational access control and credential protection for the network infrastructure.

Device R1

```
R1#show run | include hostname|secret
hostname R1
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KDfVutXOJolNq009M/
```

Figure 2: Hostname & Secret of R1

Device CSW1

```
CSW1#show run | include hostname|secret
hostname CSW1
enable secret 9 $9$J19FIAftPZf7c3$l3koe1VkzliJg5KQU.JgBhXj4RzTxEUDZnF2slb.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnxO8/0A
```

Figure 3: Hostname & Secret of CSW1

Device CSW2

```
CSW2#show run | inc host|secret
hostname CSW2
enable secret 9 $9$J19FIAftPZf7c3$l3koe1VkzliJg5KQU.JgBhXj4RzTxEUDZnF2slb.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnxO8/0A
```

Figure 4: Hostname & Secret of CSW2

Device DSW-A1

```
DSW-A1# show run | inc hostname | secret
hostname DSW-A1
enable secret 9 $9$J19FIAftPZf7c3$l3koe1VkzliJg5KQU.JgBhXj4RzTxEUDZnF2s1b.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnxO8/0A
```

Figure 5: Hostname & Secret of DSW-A1

Device DSW-A2

```
DSW-A2#show run | inc hostname| secret
hostname DSW-A2
enable secret 9 $9$J19FIAftPZf7c3$l3koe1VkzliJg5KQU.JgBhXj4RzTxEUDZnF2s1b.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnxO8/0A
```

Figure 6: Hostname & Secret of DSW-A2

Device DSW-B1

```
DSW-B1#show run | inc host|secret
hostname DSW-B1
enable secret 9 $9$J19FIAftPZf7c3$l3koe1VkzliJg5KQU.JgBhXj4RzTxEUDZnF2s1b.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnxO8/0A
```

Figure 7: Hostname & Secret of DSW-B1

Device DSW-B2

```
DSW-B2#show run | inc host|secret
hostname DSW-B2
enable secret 9 $9$J19FIAftPZf7c3$l3koe1VkzliJg5KQU.JgBhXj4RzTxEUDZnF2s1b.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnxO8/0A
```

Figure 8: Hostname & Secret of DSW-B2

Device ASW-A1

```
ASW-A1#show run | inc hostname | secret
hostname ASW-A1
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KDfVutXOJolNq009M/
```

Figure 9: Hostname & Secret of ASW-A1

Device ASW-A2

```
ASW-A2#show run | inc hostname | secret
hostname ASW-A2
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KDfVutXOJolNq009M/
```

Figure 10: Hostname & Secret of ASW-A2

Device ASW-A3

```
ASW-A3#show run | inc hostname | sec
hostname ASW-A3
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KDfVutXOJolNq009M/
```

Figure 11: Hostname & Secret of ASW-A3

Device ASW-B1

```
ASW-B1#show run | inc hostname | secret
hostname ASW-B1
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KDfVutXOJolNq009M/
```

Figure 12: Hostname & Secret of ASW-B1

Device ASW-B2

```
ASW-B2#show run | inc hostname | secret
hostname ASW-B2
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KDfVutXOJolNq009M/
```

Figure 13: Hostname & Secret of ASW-B2

Device ASW-B3

```
ASW-B3#show run | inc hostname | secret
hostname ASW-B3
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KDfVutXOJolNq009M/
```

Figure 14: Hostname & Secret of ASW-B3

2 Part 2 – VLANs, Layer-2 EtherChannel

This section covers Layer-2 EtherChannel configuration using PAgP in Office A and LACP in Office B, trunk link setup with DTP disabled and native VLAN 1000, VTPv2 implementation for VLAN propagation, creation of office-specific VLANs (PCs, Phones, Wi-Fi/Servers, Management), access port VLAN assignments, WLC1 connectivity configuration, and disabling unused ports for security.

Steps 1 & 2

```
DSW-A1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------

1      Po1(SU)           PAgP   Gig1/0/4(P) Gig1/0/5(P)
```

Figure 15: DSW-A1 - Flags & Protocols: S, U, P, PAgP & LACP

```
DSW-B1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------

1      Po1(SU)           LACP   Gig1/0/4(P) Gig1/0/5(P)
```

Figure 16: DSW-B1 - Flags & Protocols: S, U, P, PAgP & LACP

Verify steps 3a - 3d:

```
DSW-A1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1000
Gig1/0/1    on           802.1q         trunking      1000
Gig1/0/2    on           802.1q         trunking      1000
Gig1/0/3    on           802.1q         trunking      1000

Port        Vlans allowed on trunk
Po1         10,20,40,99
Gig1/0/1    10,20,40,99
Gig1/0/2    10,20,40,99
Gig1/0/3    10,20,40,99
```

Figure 17: DSW-A1 - Trunkated Ports

```
DSW-B1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1000
Gig1/0/1    on           802.1q         trunking      1000
Gig1/0/2    on           802.1q         trunking      1000
Gig1/0/3    on           802.1q         trunking      1000

Port        Vlans allowed on trunk
Po1         10,20,30,99
Gig1/0/1    10,20,30,99
Gig1/0/2    10,20,30,99
Gig1/0/3    10,20,30,99
```

Figure 18: DSW-B1 - Trunkated Ports

Verify steps 4a & 4b:

```
DSW-A1#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : JeremysITLab
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0001.6395.E700
Configuration last modified by 0.0.0.0 at 2-28-93 22:03:25
Local updater ID is 10.1.0.2 on interface Vl10 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 9
Configuration Revision            : 0
MD5 digest                        : 0x4C 0x41 0xCD 0xC1 0xA7 0xE6 0xBB 0x81
                                    0x64 0x82 0x43 0xC5 0x31 0x77 0x2A 0x67
```

Figure 19: DSW-A1 - VTP Status

```
ASW-A1#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : JeremysITLab
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0002.4AB5.E200
Configuration last modified by 0.0.0.0 at 2-28-93 22:04:25

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 0
MD5 digest                        : 0xBC 0x20 0x69 0xEE 0x58 0xC9 0x1C 0xD0
                                    0xEE 0x53 0xD6 0x6D 0xCF 0x3D 0x8E 0xBE
```

Figure 20: ASW-A1 - VTP Status

Verify steps 4a & 4b:

```
DSW-B1#show vtp stat
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : JeremysITLab
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 000C.CF76.AE00
Configuration last modified by 0.0.0.0 at 2-28-93 22:03:22
Local updater ID is 10.3.0.2 on interface Vl10 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 9
Configuration Revision            : 0
MD5 digest                        : 0xD2 0xB4 0x25 0xAE 0x9E 0x3D 0x14 0xAB
                                    0x8F 0x8D 0xF6 0xF4 0xFF 0x16 0xFB 0xFB
```

Figure 21: DSW-B1 - VTP Status

```
ASW-B1#show vtp stat
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : JeremysITLab
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0002.163D.2400
Configuration last modified by 0.0.0.0 at 2-28-93 22:03:22

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 0
MD5 digest                        : 0xD2 0xB4 0x25 0xAE 0x9E 0x3D 0x14 0xAB
                                    0x8F 0x8D 0xF6 0xF4 0xFF 0x16 0xFB 0xFB
```

Figure 22: ASW-B1 - VTP Status

Verify steps 5a - 5d:

```
ASW-A1#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   PCs                              active
20   Phones                           active
40   Wi-Fi                            active
99   Management                       active    Fa0/1
```

Figure 23: ASW-A1 - VLan Brief

Verify steps 6a - 6d:

```
ASW-B1#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
10   PCs                              active
20   Phones                           active
30   Servers                          active
99   Management                       active    Fa0/1
```

Figure 24: ASW-B1 - VLan Brief

Verify steps 7a - 7d:

```
ASW-A1#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   PCs                              active
20   Phones                           active
40   Wi-Fi                            active
99   Management                       active    Fa0/1
```

Figure 25: ASW-A1 - VLan Brief

```
ASW-A2#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
10   PCs                              active    Fa0/1
20   Phones                           active    Fa0/1
40   Wi-Fi                            active
99   Management                       active
```

Figure 26: ASW-B2 - VLan Brief

```
ASW-B3#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
10   PCs                              active
20   Phones                           active
30   Servers                          active    Fa0/1
99   Management                       active
```

Figure 27: ASW-B3 - VLan Brief

Verify steps 8a - 8c:

```
ASW-A1#show int stat
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    99         a-full  a-100 10/100BaseTX
Fa0/2                        connected    trunk      a-full  a-100 10/100BaseTX
Fa0/3                        disabled 1          auto    auto  10/100BaseTX
Fa0/4                        disabled 1          auto    auto  10/100BaseTX
Fa0/5                        disabled 1          auto    auto  10/100BaseTX
Fa0/6                        disabled 1          auto    auto  10/100BaseTX
Fa0/7                        disabled 1          auto    auto  10/100BaseTX
```

Figure 28: ASW-A1 - Trunked Ports

```
ASW-A1#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
10   PCs                              active
20   Phones                           active
40   Wi-Fi                            active
99   Management                       active    Fa0/1
```

Figure 29: ASW-A1 - VLan Brief

Verify step 9:

```
DSW-A1#show int stat
Port      Name               Status       Vlan       Duplex  Speed Type
Po1                          connected    trunk      a-full  a-100
Gig1/0/1                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/2                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/3                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/4                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/5                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/6                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/7                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/8                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/9                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/10                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/11                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/12                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/13                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/14                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/15                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/16                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/17                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/18                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/19                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/20                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/21                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/22                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/23                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/24                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/1/1                     connected    routed     a-full  a-100 10/100/1000BaseTX
Gig1/1/2                     connected    routed     a-full  a-100 10/100/1000BaseTX
Gig1/1/3                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/1/4                     disabled 1          auto    auto  10/100/1000BaseTX
```

Figure 30: DSW-A1 - All Ports Status

```
DSW-A2#show int stat
Port      Name               Status       Vlan       Duplex  Speed Type
Po1                          connected    trunk      a-full  a-100
Gig1/0/1                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/2                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/3                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/4                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/5                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/6                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/7                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/8                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/9                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/10                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/11                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/12                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/13                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/14                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/15                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/16                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/17                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/18                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/19                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/20                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/21                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/22                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/23                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/24                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/1/1                     connected    routed     a-full  a-100 10/100/1000BaseTX
Gig1/1/2                     connected    routed     a-full  a-100 10/100/1000BaseTX
Gig1/1/3                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/1/4                     disabled 1          auto    auto  10/100/1000BaseTX
```

Figure 31: DSW-A2 - All Ports Status

Verify step 9:

```
DSW-B1#show int stat
Port      Name               Status       Vlan       Duplex  Speed Type
Po1                          connected    trunk      a-full  a-100
Gig1/0/1                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/2                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/3                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/4                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/5                     connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig1/0/6                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/7                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/8                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/9                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/10                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/11                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/12                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/13                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/14                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/15                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/16                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/17                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/18                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/19                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/20                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/21                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/22                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/23                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/0/24                    disabled 1          auto    auto  10/100/1000BaseTX
Gig1/1/1                     connected    routed     a-full  a-100 10/100/1000BaseTX
Gig1/1/2                     connected    routed     a-full  a-100 10/100/1000BaseTX
Gig1/1/3                     disabled 1          auto    auto  10/100/1000BaseTX
Gig1/1/4                     disabled 1          auto    auto  10/100/1000BaseTX
```

Figure 32: DSW-B1 - All Ports Status

```
ASW-B2#show int stat
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        disabled 1          auto    auto  10/100BaseTX
Fa0/3                        disabled 1          auto    auto  10/100BaseTX
Fa0/4                        disabled 1          auto    auto  10/100BaseTX
Fa0/5                        disabled 1          auto    auto  10/100BaseTX
Fa0/6                        disabled 1          auto    auto  10/100BaseTX
Fa0/7                        disabled 1          auto    auto  10/100BaseTX
Fa0/8                        disabled 1          auto    auto  10/100BaseTX
Fa0/9                        disabled 1          auto    auto  10/100BaseTX
Fa0/10                       disabled 1          auto    auto  10/100BaseTX
Fa0/11                       disabled 1          auto    auto  10/100BaseTX
Fa0/12                       disabled 1          auto    auto  10/100BaseTX
Fa0/13                       disabled 1          auto    auto  10/100BaseTX
Fa0/14                       disabled 1          auto    auto  10/100BaseTX
Fa0/15                       disabled 1          auto    auto  10/100BaseTX
Fa0/16                       disabled 1          auto    auto  10/100BaseTX
Fa0/17                       disabled 1          auto    auto  10/100BaseTX
Fa0/18                       disabled 1          auto    auto  10/100BaseTX
Fa0/19                       disabled 1          auto    auto  10/100BaseTX
Fa0/20                       disabled 1          auto    auto  10/100BaseTX
Fa0/21                       disabled 1          auto    auto  10/100BaseTX
Fa0/22                       disabled 1          auto    auto  10/100BaseTX
Fa0/23                       disabled 1          auto    auto  10/100BaseTX
Fa0/24                       disabled 1          auto    auto  10/100BaseTX
Gig0/1                       connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig0/2                       connected    trunk      a-full  a-100 10/100/1000BaseTX
```

Figure 33: ASW-B2 - All Ports Status